



GE Medical Systems

gehealthcare.com

Technical Publication

Direction 2388683-100

Revision 02

GE Medical Systems

UPS Installation: LightSpeed 5.X

For use with NGPDU models:

- 2326492
- 2326492-3
- 2326492-4



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- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
1	04/20/04	Initial Release
2	12/20/04	PQR 13031029 - improved legibility of drawings in Chapter 6. Added Warning (Use Correct UPS) to Chapter 3.

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Chapter 1

Common Instructions and Requirements

Section 1.0

Scope

This installation specification is a control document for the installation of an Uninterruptible Power Supply (UPS) unit to be used by General Electric Medical Systems (GEMS) in the Computerized Tomography diagnostic imaging system. This document supports CT Systems that use the following Power Distribution Units.

NGPDU models:

- 2326492
- 2326492-3
- 2326492-4

It also includes the instructions for installation of a UPS Disconnect Panel and interconnection with the system A1 Main Disconnect & UPS Control Panel.

Section 2.0

Overview

This document was designed to allow installation of a 12.5kVA UPS into LightSpeed 5.x systems using an NGPDU. The UPS primarily backs up the system electronics and computer functions. It is intended to bridge short power outages and provide time for crossover from normal mains power to emergency power. It is sized to provide at least 10 minutes backup with fully charged batteries. It is recommended that if a power outage, with no mains or emergency power, exceeds 5 minutes, the patient should be removed from the table and a shutdown of the system should be initiated.

A UPS Disconnect Panel is installed with the UPS unit to provide electrical disconnect of the UPS from the CT System. It also provides a contact closure that shuts the UPS down when system power is turned off by way of any of the emergency off switches.

For safe and orderly system start-up & shutdown this UPS kit must be installed with a properly configured A1 Main Disconnect Panel. Use of one of the following GE panels specifically designed for use with this kit, is recommended:

- E4502AB (90A)
- E4502AC (110A)
- E4502AE (125A)
- E4502AF (150A)

Call Powerware (800-843-9433). Have your UPS S/N ready when you call.

2.1 Tools Required

- FE Tool Kit
- LOTO
- Fluke 87 Meter
- Wire Stripper
- Labels

2.2 Time

3 hours

2.3 Manpower

- 1 FE
- 1 Electrician

Section 3.0

Safety Requirements - General Warning Considerations

The only user operations permitted with the UPS are:

- Starting up and shutting down of the UPS.
- Operating the user interface in the UPS.
- Connecting data interface cables.
- Monitoring the UPS.

Note: It is strongly recommended to have the User's Guide for the UPS available. These operations must be performed according to the instructions in this document and the User's Guide for the UPS being installed. This installation document covers the POWERWARE 9150 UPS only.



DANGER

THE UPS CONTAINS LETHAL VOLTAGES. SERVICE AND REPAIRS SHOULD BE PERFORMED BY AUTHORIZED PERSONNEL ONLY. THERE ARE NO USER SERVICEABLE PARTS INSIDE THE UPS.

Section 4.0

Preliminary Instructions

4.1 Electrician

An electrician will be required to install a 1/2" conduit between the A1 Disconnect and the computer floor or to the general area of the UPS. The conduit should be terminated in a small wall box. This will be used for the UPS Control cable connecting the UPS Disconnect Panel and A1 Main Disconnect Panel.

4.2 Uncrating

Locate the uncrating instructions for the UPS found on or in the UPS packing. Uncrate the UPS per these instructions. Visually inspect the unit for shipping damage and battery seating. The POWERWARE 9150 model requires assembly of the subsystem. Accomplish this per drawing 110719345 in the POWERWARE 9150 (Physical manual supplied with the unit).

4.3 Electrical Requirements

The UPS receives power from the NGPDU and returns power to the NGPDU via the UPS Disconnect Panel. No other power connections are required.

Chapter 2

Powerware 9150 UPS Installation

Perform the following installations and connections at the UPS and UPS Disconnect Panel.

Section 1.0

Powerware 9150 UPS Mechanical Preparation

1.1 UPS Cover Removal

- 1.) Remove the front and left side covers from the POWERWARE 9150 UPS as follows:
 - a.) The front panel is held in place by spring clips. To remove the front panel, pull front panel forward using the “Hand hold” provided on the lower right side of the front panel.
 - b.) Two screws (top and bottom) should be removed from the lower left corner of the assembly to release the Left Panel.
 - c.) The left side panel is removed by pulling the front edge of the left panel outward and then by sliding the panel towards the back of the UPS.

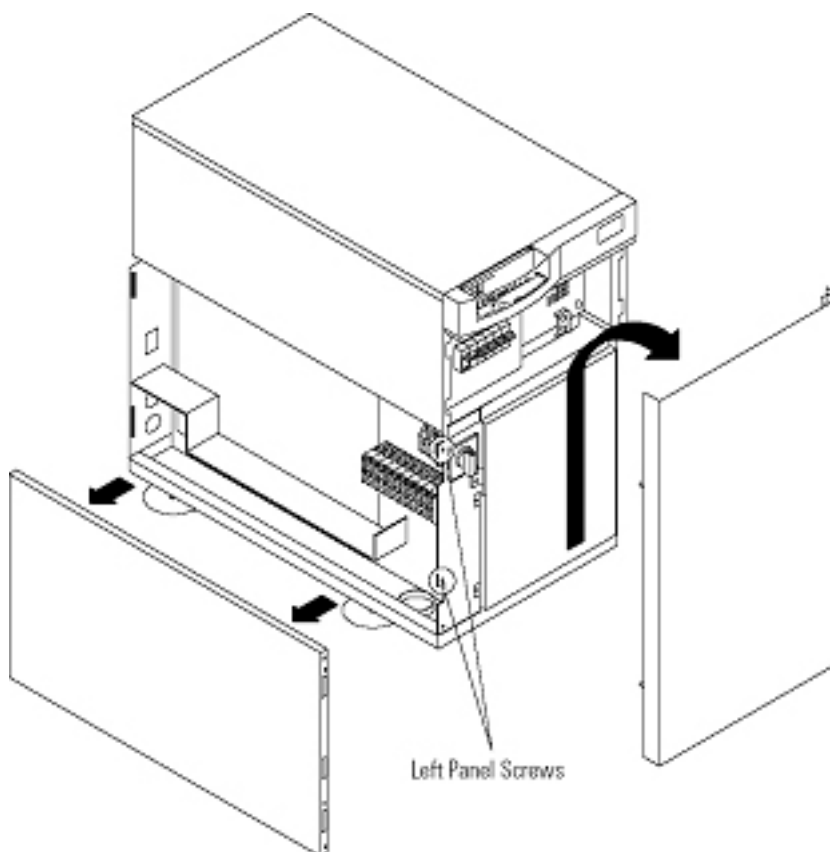


Figure 2-1 Removing the UPS Panels

- 2.) Verify that all of the battery breakers on front of UPS are in the “Off” position.
- 3.) Confirm by measurement that all terminals on terminal strip are at ground potential. TB1

- 4.) Install one each 3/4", & 7/8" cord grip fittings in rear of UPS as shown in [Figure 2-2](#). Use reducer plates and hardware supplied in the kit.

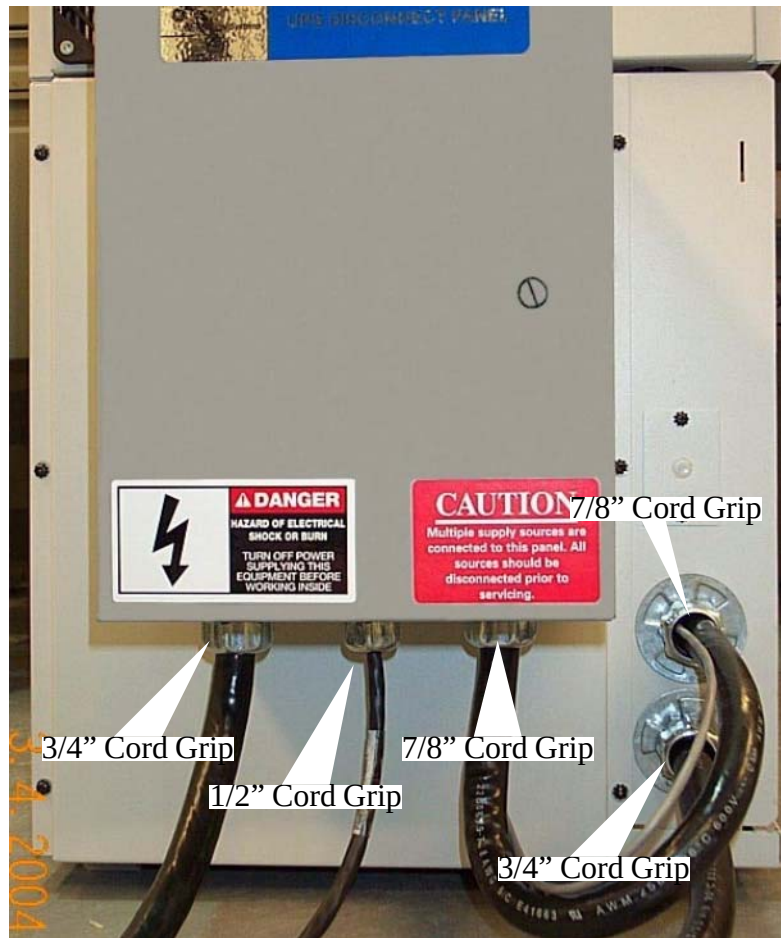


Figure 2-2 UPS Rear View

1.2 UPS Disconnect Panel

The UPS Disconnect Panel may be mounted in either of two locations.

1.2.1 UPS MOUNT

The UPS Disconnect Panel may be mounted on the rear of the UPS as shown in [Figure 2-2](#) and described below.

- 1.) Remove and discard the UPS lower rear cover panel.
- 2.) Retain six (6) mounting screws and use to install replacement rear panel from kit.
- 3.) Install one each 1/2", 3/4", & 7/8" cord grip fittings in bottom of disconnect box as shown in [Figure 2-2](#), using mounting hardware supplied in the kit.
- 4.) Mount the UPS Disconnect Panel on the UPS rear cover panel, using four (4) mounting studs and hardware supplied in the kit.

1.2.2 WALL MOUNT

If local code permits, the UPS Disconnect Panel may be wall mounted within 10 feet (3.5 meters) of the UPS location. When wall mounted the connecting cables should be protected by raceway or conduit. Mechanical details of the wall mount option are left to the installer.

Section 2.0

Powerware 9150 UPS Connections

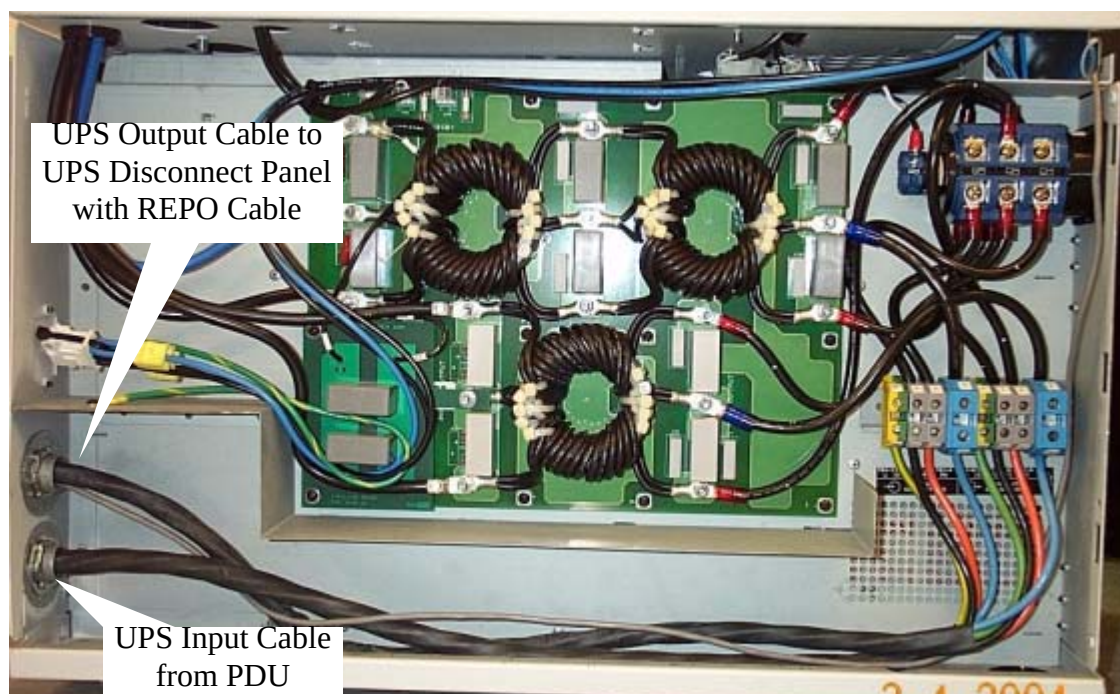


Figure 2-3 UPS Side View

2.1 UPS Input Power Cable from PDU

Connect UPS Input Power Cable (P/N 2391751) as follows:

- 1.) Route cable end labeled "To UPS TB1" through the 3/4" cord grip in the back of the UPS, to TB1.
- 2.) Connect the GREEN/YELLOW wire to TB1-1 (GND).
- 3.) Connect the BLACK wire to TB1-2 (L1).
- 4.) Connect the ORANGE wire to TB1-3 (L2).
- 5.) Connect the BLUE wire to TB1-4 (NEUT).
- 6.) If present, cut off RED wire at cable jacket.
- 7.) Securely tighten the cord grip on the input cable.



Figure 2-4 UPS TB1 Connections

2.2 UPS Output Cable to UPS Disconnect Panel

Connect UPS Output Power Cable (P/N 2391751-2) as follows:

- 1.) Route cable end labeled "To UPS TB1" through the 7/8" cord grip in the back of the UPS, to TB1.
- 2.) Connect the GREEN/YELLOW wire to TB1-6 (GND).
- 3.) Connect the BLACK wire to TB1-7 (L1).
- 4.) Connect the ORANGE wire to TB1-8 (L2).
- 5.) Connect the BLUE wire to TB1-9 (NEUT).
- 6.) If present, cut off RED wire at cable jacket.

2.3 UPS Internal Interlock (REPO)

Connect the UPS REPO interlock as follows:

- 1.) Locate the 2-pin male connector (REPO) plugged into the bottom side of the upper front panel. See [Figure 2-5](#) below. Unplug the connector. Remove the jumper that connects the two pins.

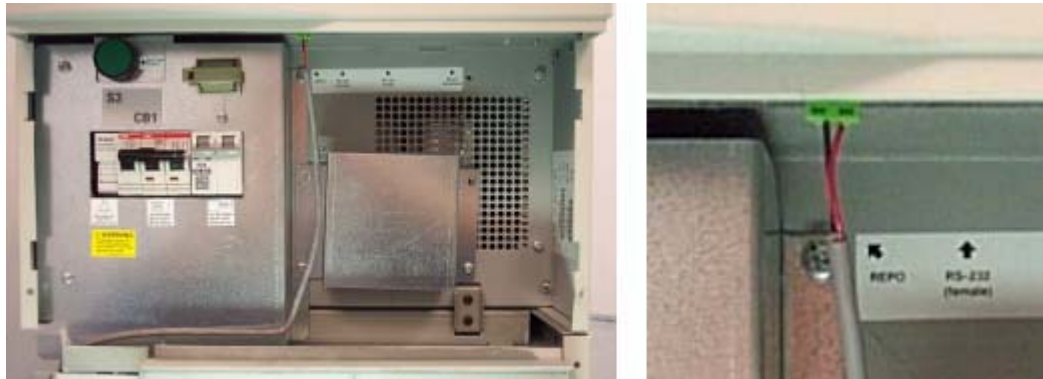


Figure 2-5 REPO Cable Connections

- 2.) Locate the REPO Cable (P/N 2315458) in the kit and connect the RED (#22) and BLACK (#22) wires labeled "REPO-A" & "REPO-B" to the two terminals of the connector.
- 3.) Replace the connector in its socket.
- 4.) Dress the REPO cable down to the TB1 area so that covers will not pinch them.
- 5.) Dress the REPO cable with the output cable along the lower section of the UPS side chamber and exit out the rear through the 7/8" cord grip. Note: both cables will pass through the same cord grip. Securely tighten the cord grip.
- 6.) Replace UPS left side cover and secure with two screws.

Section 3.0

UPS Disconnect Panel Connections

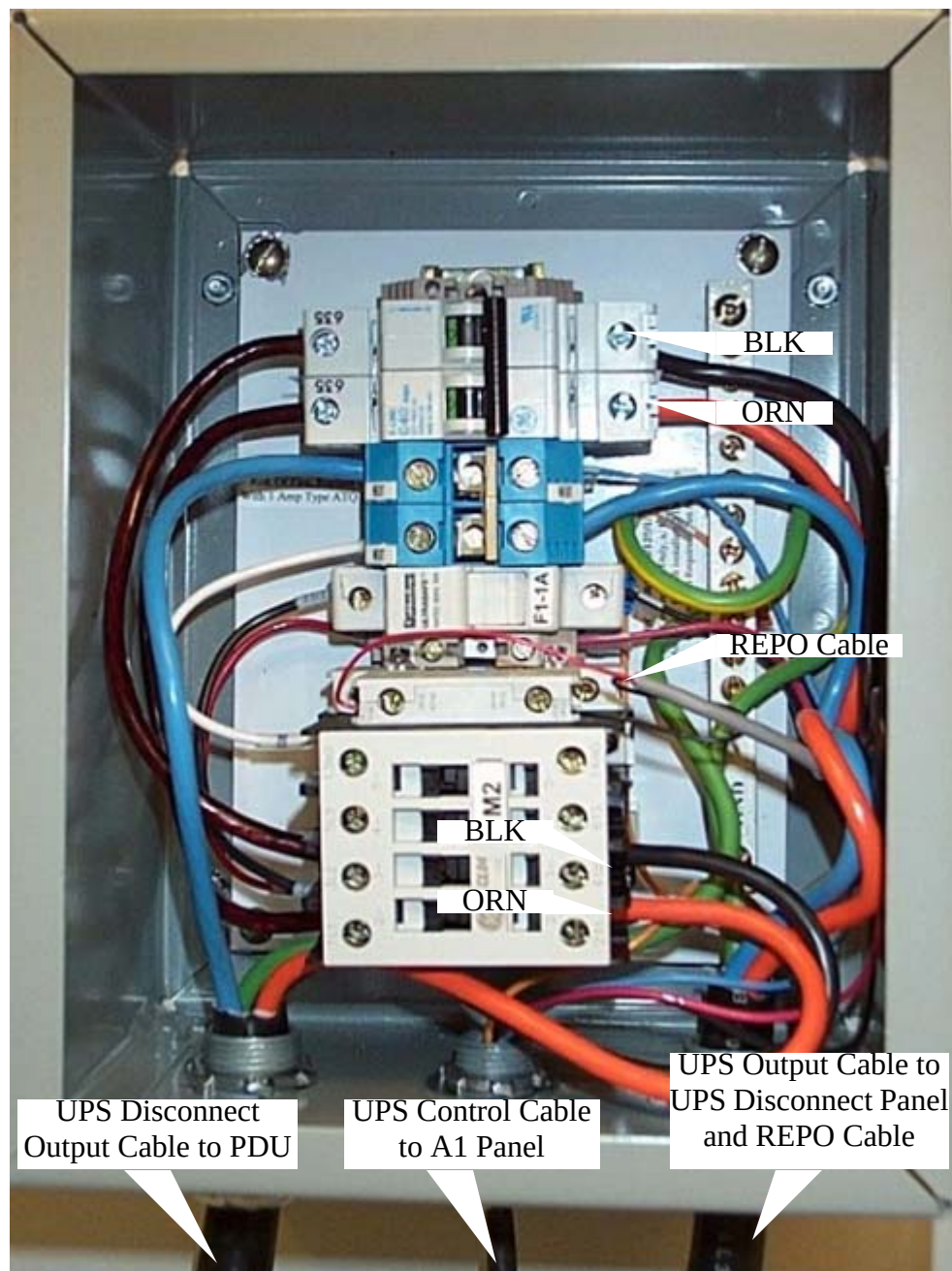


Figure 2-6 Figure 5 - UPS Disconnect Panel Connections

3.1 UPS Output Cable & REPO Cable to UPS Disconnect Panel

Depending on the location chosen for the UPS Disconnect Panel, cut the UPS Output cable (P/N 2391751-2) and REPO cable (P/N 2315458) as required. Re-terminate the wires with ferrules provided in the kit. Connect the two cables as follows:

- 1.) Dress the UPS Output and REPO cables neatly and pull the ends initially labeled "To UPS DISC PNL" through the 7/8" cord grip in the bottom of the UPS Disconnect Panel.
- 2.) Connect the GREEN/YELLOW (#8) wire to the Grounding bar "GND".
- 3.) Connect the BLACK (#8) wire to CB1-A (upper terminal).
- 4.) Connect the ORANGE (#8) wire to CB1-B (lower terminal).
- 5.) Connect the BLUE (#8) wire to the blue "NEUT" block.
- 6.) If present, cut off RED (#8) wire at cable jacket.
- 7.) Connect the BLACK (#22) wire from the REPO cable to the "44NO" terminal on the M2 contactor.
- 8.) Connect the RED (#22) wire from the REPO cable to "43NO" terminal on the M2 contactor.
- 9.) Securely tighten the cord grip onto the cable(s).

3.2 UPS Disconnect Output Cable to PDU

Connect UPS Disconnect Output cable (P/N 2391751-3) as follows:

- 1.) Route the cable end labeled "To UPS DISC PNL" through the 3/4" cord grip in the bottom of the UPS Disconnect Panel.
- 2.) Connect the GREEN/YELLOW (#8) wire to the Grounding bar "GND".
- 3.) Connect the BLACK (#8) wire to terminal 4T2 on the M2 contactor.
- 4.) Connect the BLUE (#8) wire to the blue "NEUT" block.
- 5.) Connect the ORANGE (#8) wire to terminal 2T1 on the M2 contractor.
- 6.) If present, cut off RED (#8) wire at cable jacket.
- 7.) Securely tighten the cord grip onto the cable.

3.3 UPS Control Cable to A1 Main Disconnect / UPS Control Panel

Connect UPS Control cable (P/N 2396155) as follows:

- 1.) Route the cable end labeled "To UPS DISC PNL" through the 1/2" cord grip in the bottom of the UPS Disconnect Panel.
- 2.) Connect the GREEN/YELLOW (#18) wire to the Grounding bar "GND". (May be different color wire with ground indicator.)
- 3.) Connect the BLACK (#18) wire to Fuse "F1A".
- 4.) Connect the BLUE (#18) wire to the blue "NEUT" block.
- 5.) Connect the RED (#18) wire to the terminal labeled "COIL".
- 6.) Securely tighten the cord grip onto the cable.

Chapter 3

NGPDU - Power Distribution Unit

This UPS kit has been specifically designed for compatibility with LightSpeed 5.x series CT systems. These systems utilize an NGPDU. Model numbers currently in use are:

- 2326492
- 2326492-3
- 2326492-4

The same interconnection information applies to all three listed NGPDU models.

Section 1.0 Shutdown

- 1.) Shutdown system software.
- 2.) Turn off the mains input power to the system.
- 3.) LOTO.



WARNING

**POTENTIAL FOR ELECTRIC SHOCK.
USE PROPER LOCK-OUT/TAG-OUT SAFETY PROCEDURES.**



WARNING

**UPS FUNCTION FAILURE, EQUIPMENT DAMAGE OR ELECTRIC SHOCK TO PERSONNEL IS POSSIBLE, DUE TO WRONG UPS TYPE OR INCORRECT CONNECTION OF UPS.
CHOOSE THE DEFINED UPS TYPE AND FOLLOW THE CONNECTION INSTRUCTION STRICTLY.**

Section 2.0 UPS Input Power Cable

- 1.) Remove the top and front covers of the PDU.
- 2.) Remove all three jumper wires at terminal strip TS4, terminals 1 to 6, 2 to 7, & 3 to 8. Discard the jumpers.

- 3.) Route the UPS Input Power Cable (P/N 2391751) into the PDU and up through the indicated cable clamp in the bottom pan. Refer to [Figure 3-1](#) below.

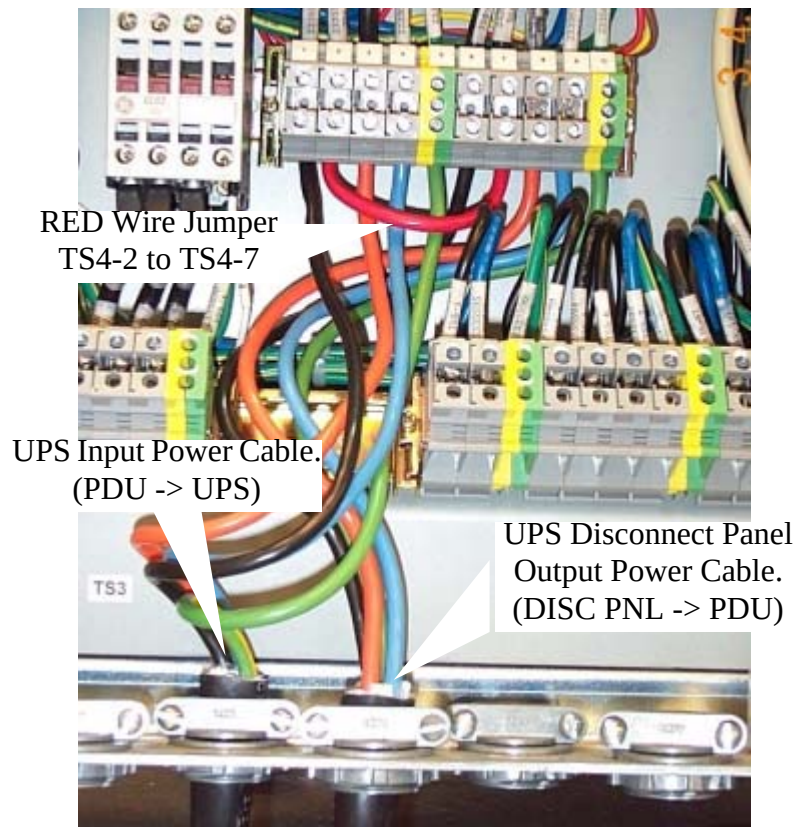


Figure 3-1 Figure 6 - NGPDU Connections

- 4.) Cut off the RED wire at the cable jacket and set aside the cutoff piece.
- 5.) Connect the BLACK wire to TS4-1
- 6.) Connect the ORANGE wire to TS4-3.
- 7.) Connect the BLUE wire to TS4-4.
- 8.) Connect the GREEN/YELLOW wire to TS4-5.
- 9.) Using the RED wire set aside above, cut to length required and install as a jumper between TS4-2 and TS4-7.

Section 3.0

UPS Disconnect Panel Output Power Cable

- 1.) Route the UPS Disconnect Panel Output Power Cable (P/N 2391751-3) into the PDU and up through the indicated cable clamp in the bottom pan.
- 2.) Cut off the RED wire at the cable jacket and discard.
- 3.) Connect the BLACK wire to TS4-6
- 4.) Connect the ORANGE wire to TS4-8.
- 5.) Connect the BLUE wire to TS4-9.
- 6.) Connect the GREEN/YELLOW wire to TS4-10

Note: Do not re-install the PDU covers at this time. Wait until after the system start-up procedures have been completed.

Chapter 4

A1 Disconnect Panel

Section 1.0

A1 Main Disconnect Panel Description

For safe and orderly system start-up & shutdown this UPS kit must be installed with a properly configured A1 Main Disconnect Panel. Use of one of the following GE panels specifically designed for use with this kit, is recommended:

- E4502AB (90A)
- E4502AC (110A)
- E4502AE (125A)
- E4502AF (150A)

See the appropriate Pre-Install manual for your system.



Figure 4-1 Typical A1 Panel Front View & Interior View



WARNING

PULL THE MANUAL HANDLE DOWN ON THE A1 DISCONNECT PANEL TO DISCONNECT THE MAINS INPUT FROM SYSTEM. USE PROPER LOCK-OUT/TAG-OUT PROCEDURES. THE FOLLOWING PROCEDURES ARE ACCOMPLISHED INSIDE THE A1 DISCONNECT PANEL.



DANGER

MAINS VOLTAGE IS STILL PRESENT INSIDE THE A1 PANEL WITH MANUAL SWITCH PULLED. LOCATE THE FEEDER DISCONNECT TO THIS PANEL AND SHUT MAIN POWER OFF THERE. USE PROPER LOCK-OUT/TAG-OUT PROCEDURES. TEST THE A1 DISCONNECT TO VERIFY THAT ALL POWER IS OFF.

Section 2.0

UPS Control Cable Connections to A1 Panel

An electrician will be required to install a 1/2" conduit or raceway between the A1 Disconnect and the floor in the general area of the UPS. Conduit should be terminated in a small wall box. This route will be used for the UPS Control cable connecting the UPS Disconnect Panel and A1 Main Disconnect Panel.

- 1.) Install a 1/2" cord grip (supplied with the kit) in the wall box or the base of the A1 Panel.
- 2.) Route the UPS Control Cable (P/N 2396155) from the UPS Disconnect Panel, through the conduit or raceway to the A1 Disconnect Panel.
- 3.) Connect the BLACK wire to terminal TB1-9 in the A1 Panel.
- 4.) Connect the RED wire to terminal TB1-8 in the A1 Panel.
- 5.) Connect the BLU wire to terminal TB1-10 in the A1 Panel.
- 6.) Connect the GREEN/YELLOW (could also be orange) wire to terminal TB1-11 (GND) in the A1 Panel.

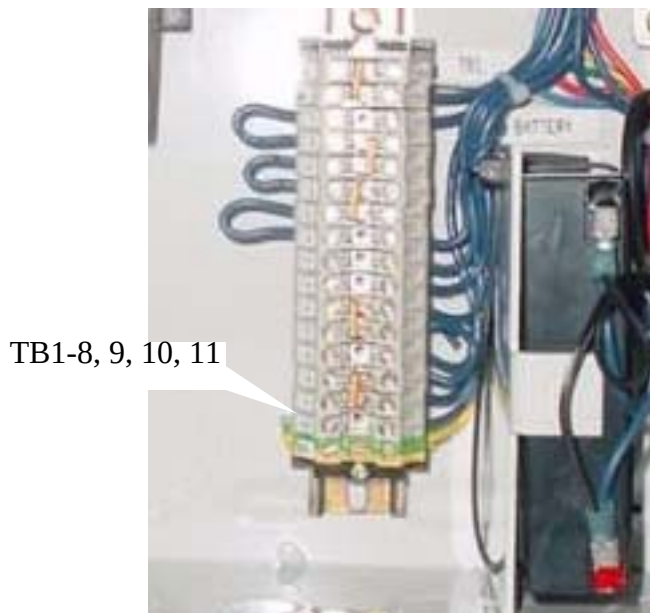


Figure 4-2 A1 Panel TB1 Terminals

Section 3.0

Add Label

- 1.) A1 Panel - Add Startup Label
- 2.) UPS - Add Startup Label

Chapter 5

Start Up / Shutdown Procedures

Use this procedure to bring the system up the first time power is applied after installing the UPS kit. Verify that all connections are properly made and all safety covers are properly in place. Leave the PDU front cover off. Turn OFF the Console power switch. This keeps the system computers from needlessly booting up until proper UPS system operation is verified. Note: It is strongly recommended to have the User's Guide for the particular UPS been used.

Check the following before proceeding to power up the UPS unit:

- Mains input power is OFF. Restore the power at the feeder panel supplying the A1 Disconnect Panel after finishing the procedure.
- **STOP.** Call Powerware (800-843-9433). Have your UPS S/N ready when you call.

Section 1.0

Start-Up: POWERWARE 9150

- 1.) Set the UPS On-Off toggle switch on its front control panel (under plastic cover) to the "OFF" position.
- 2.) Remove the UPS front cover panel and turn the Maintenance Bypass switch to the "SERVICE" position.
- 3.) Set battery circuit breaker(s) CB1 to "ON".
- 4.) Verify the CT System is ready to receive power and all personnel are clear of the machine.
- 5.) Set the manual disconnect switch in the A1 Main Disconnect Panel to the "ON" position and turn the Key Switch to the "ON" position, if required.
- 6.) Momentarily press the "START" button on the A1 Main Disconnect Panel. Verify the "UPS ON" and "Contactor Power" lights turn on and the system begins to power up.

Note: If the system fails to start up or the main contactor does not stay energized, check the following:

- Verify no system SEO is locked open.
 - Verify CB1 and F1 in the UPS Disconnect Panel are OK.
 - Verify all fuses in the A1 Disconnect Panel are OK.
 - Verify the PDU circuit breaker(s) are OK.
- 7.) Start the UPS by setting the UPS On-Off toggle switch to the "ON" position.
 - 8.) The UPS should begin its internal startup sequence and synchronize to the utility power line in "Automatic Configuration Mode". It will beep periodically while it completes internal startup. After about 3 to 5 minutes it should display the UPS ON, LINE ON, BYPASSED and LOAD indicators. (See Section 1.3 for Beep Codes.)
 - 9.) Turn the Maintenance Bypass Switch fully counter-clockwise to the "UPS" position (normal position).
 - 10.) After approximately 30sec, the BYPASSED indicator should turn off when the power is transferred to the inverter.
 - 11.) Reinstall the front cover of the UPS and reinstall the PDU covers. Turn ON the Console power switch.

Note: The UPS is now providing 120VAC power to the Table, Gantry and Console. In the event of a momentary mains power loss, scanning will be aborted / inhibited. However, the UPS will continue

to provide power to the Table, Gantry & Console for at least 10 minutes. The UPS will “beep” occasionally, letting you know that it is on BATTERY BACKUP. If mains power is restored prior to shutting down the CT system, the UPS and CT system will automatically return to normal operation.

1.1 Automatic Configuration Mode

Automatic configuration is performed during UPS startup. The UPS attempts to match its output voltage and frequency to the input power. Once the UPS determines the input voltage and frequency it beeps twice and sequences through all the LEDs. One of the LEDs will remain blinking. The blinking LED shows the configuration, which has been automatically selected (see the configuration table below).

1.1.1 60 Hz Systems

If the automatic selection is the same as the latest previously used (reference) configuration, the LED will blink for only 10 seconds, the sequencing LEDs will be displayed again and the UPS will start.

1.1.2 50 Hz Systems

The UPS is tested at the factory in a 120/208Vac, 60Hz mode. When it is installed on a 50 Hz system, the UPS will sense the change in frequency on first start-up. Whenever the sensed input power voltage or frequency differs from the latest previously used (reference) configuration setting, an LED will blink indicating the automatic configuration selection as shown in the table below. However, the UPS will not start until the RESET push-button is pushed. If the selection is correct, push the RESET button momentarily to store this selection in the UPS.

On subsequent start-up of the UPS it will automatically select the correct 50 Hz output.

1.1.3 Correct Configuration - 120/208V



NOTICE Verify the selection is 120/208V, 50 or 60 Hz, depending on input frequency.

Once the output selection has been confirmed the UPS beeps twice and sequences the LEDs. The LED indicating the selection blinks for 10 seconds and the LEDs sequence once again and the UPS starts. If required, refer to the UPS User's Guide for further information. Acceptable selections are shown in **BOLD** below.

Blinking LED	Input/Output Voltage Configuration	Output Frequency
ALARM	120/208V	50 Hz
SERVICE	110/220V	50 Hz
OVERTEMP	127/220V	50 Hz
OVERLOAD	100/200V	50 Hz
80%	120/240V	50 Hz
60%	120/208V	60 Hz
40%	110/220V	60 Hz
LOAD	127/220V	60 Hz
BYPASSED	100/200V	60 Hz
ON BATTERY	120/240V	60 Hz

Table 5-1 Blinking Configuration

1.2 UPS Alarm Conditions

1.2.1 Input Phase Reversal

If the UPS does not start and the ALARM LED is On or Blinking (11 short beeps and 5 long beeps), there is an input phase error. Two of the main input phases must be rotated at the PDU MAINS INPUT Terminal Block. Do NOT reverse any of the connections at the UPS or PDU TS4 Interface.

1.2.2 Other Alarms

Refer to the UPS User's Guide for a detailed listing of diagnostic information for the UPS.

Section 2.0 Shut-Down Procedure

2.1 System Shutdown

To shut down the entire system, do the following:

- 1.) Perform a software shutdown.
- 2.) Turn off the Console power switch.
- 3.) Press any room System Emergency Off (SEO) button or
- 4.) Press the STOP button on the A1 Main Disconnect Panel
- 5.) Set the manual switch of the A1 Disconnect Panel in the OFF position.

To re-start the system, follow the preceding Start-Up instructions.

2.2 Powerware 9150 Only

To shut down the UPS only while leaving the system operational do the following:

- 1.) Remove the UPS front panel.
- 2.) Turn the Maintenance Bypass switch to the BYPASS position.
- 3.) Turn "OFF" the UPS ON/OFF toggle switch (under the plastic cover).
- 4.) Turn the UPS battery circuit breaker CB1 and CB2 to the "OFF" position.
- 5.) Reinstall the UPS front panel.



CAUTION

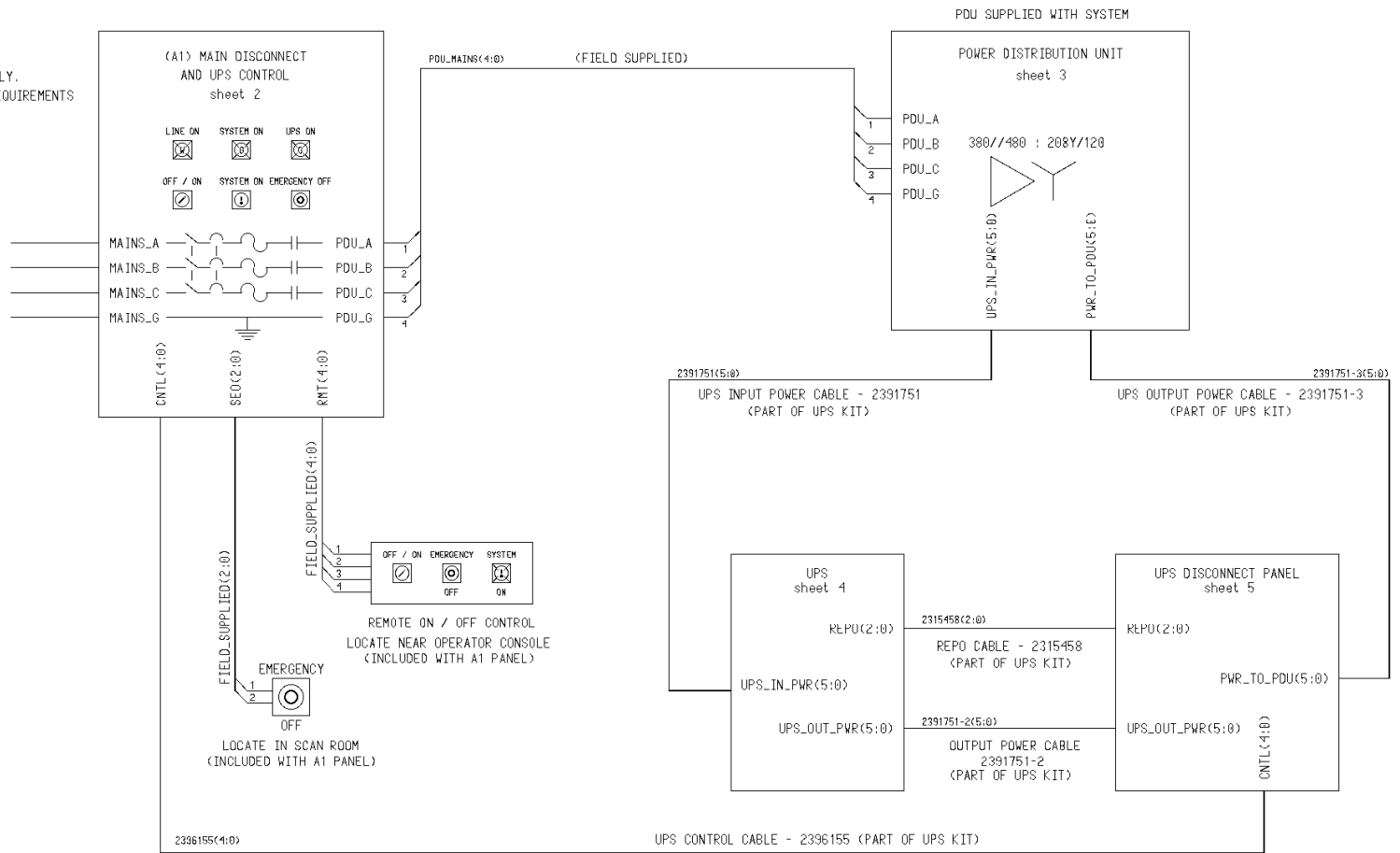
Although the UPS stops supplying power and is disconnected internally from the batteries, system power still passes through the UPS cables. Use caution when servicing.

Chapter 6

Drawings

NOTE: A1 PANEL SHOWN FOR INFORMATION ONLY.
PURCHASE SEPARATELY PER SYSTEM REQUIREMENTS
EXAMPLE: E4502AB - 90A
E4502AC - 110A
E4502AE - 125A
E4502AF - 150A

CUSTOMER MAINS
INPUT TO SYSTEM



SET_NAME	SHT NUM	REV
A1 DTSC	2	A

Figure 6-1 2391780 SCH, Sheet 1, Rev 0

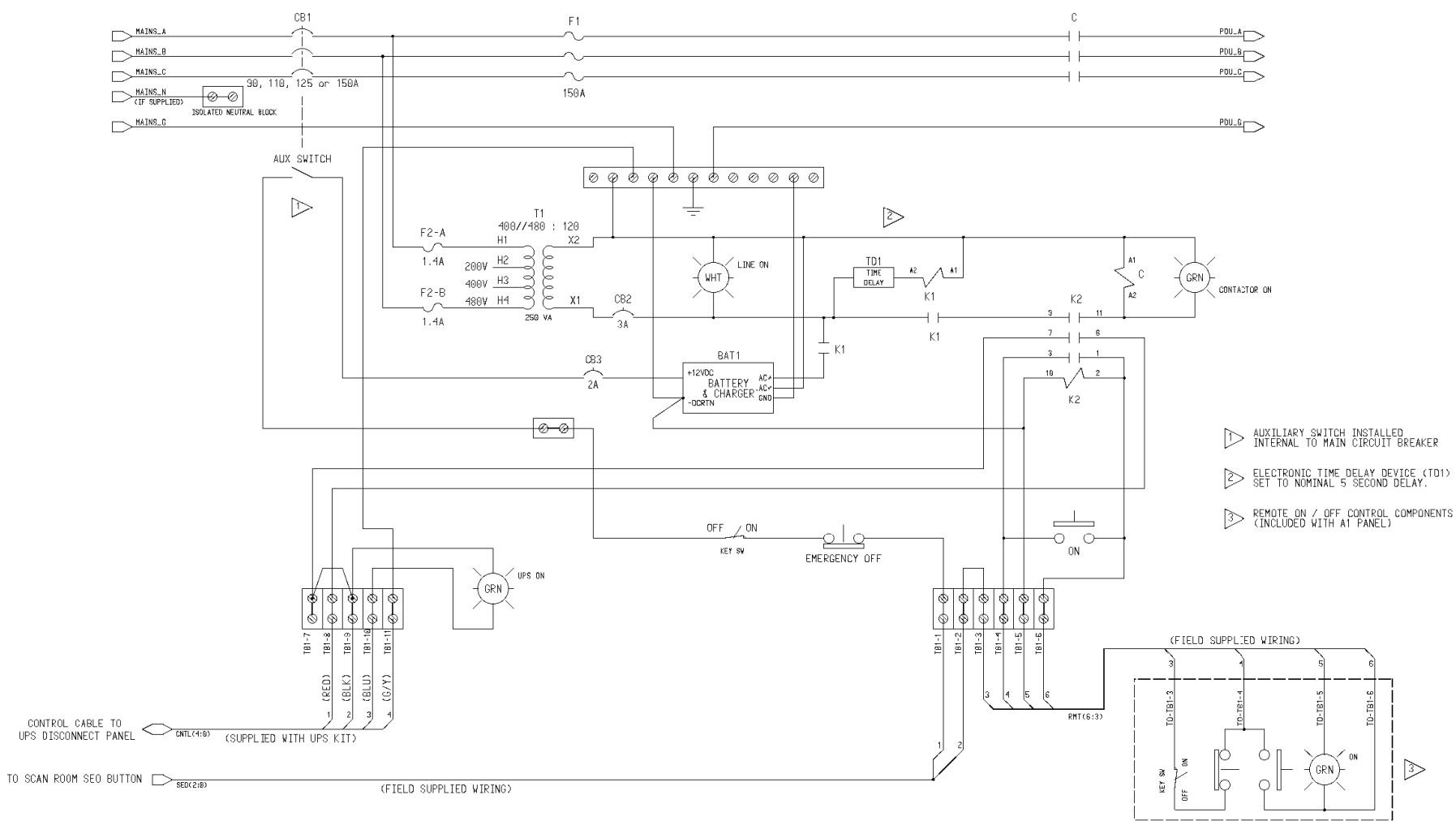
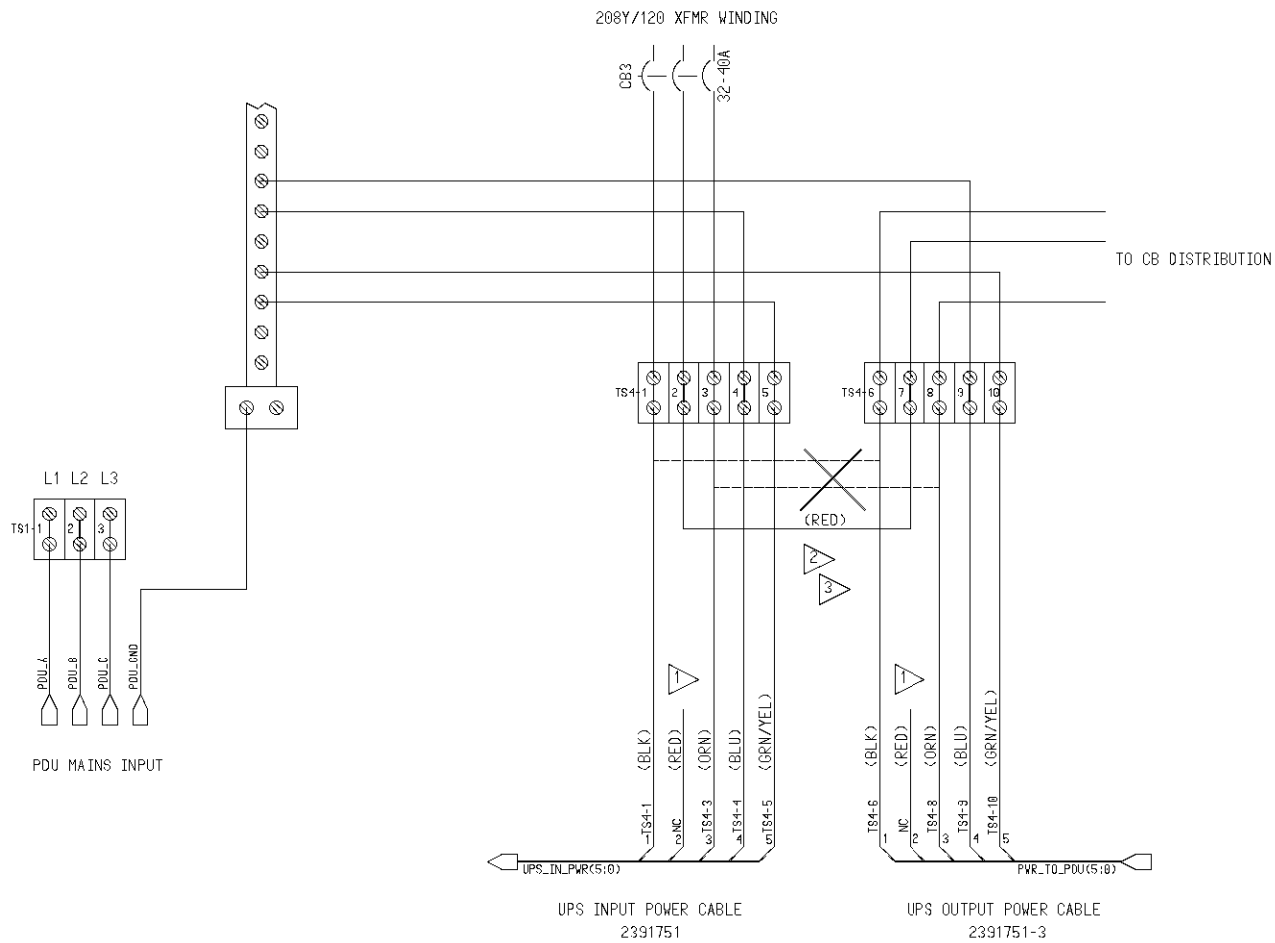


Figure 6-2 2391780 SCH, Sheet 2, Rev 0



- 1 RED WIRES IN CABLES ARE NOT USED CUT OFF AT CABLE JACKET.
- 2 REMOVE FACTORY INSTALLED JUMPERS BETWEEN TS4-1/2/3 AND TS4-6/7/8
- 3 USE RED WIRE CUT FROM CABLE TO REPLACE JUMPER BETWEEN TS4-2 AND TS4-7.

Figure 6-3 2391780 SCH, Sheet 3, Rev 0

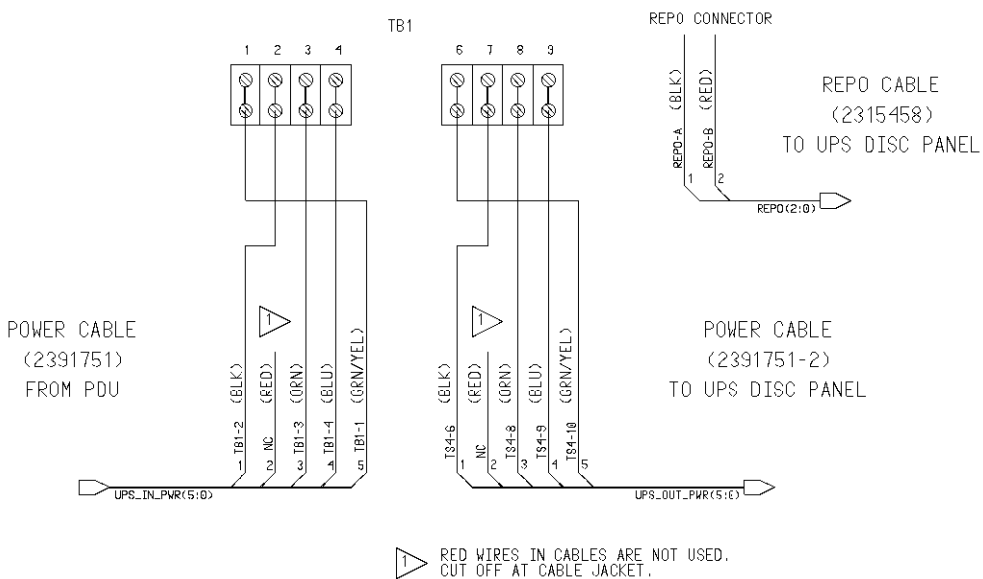


Figure 6-4 2391780 SCH, Sheet 4, Rev 0

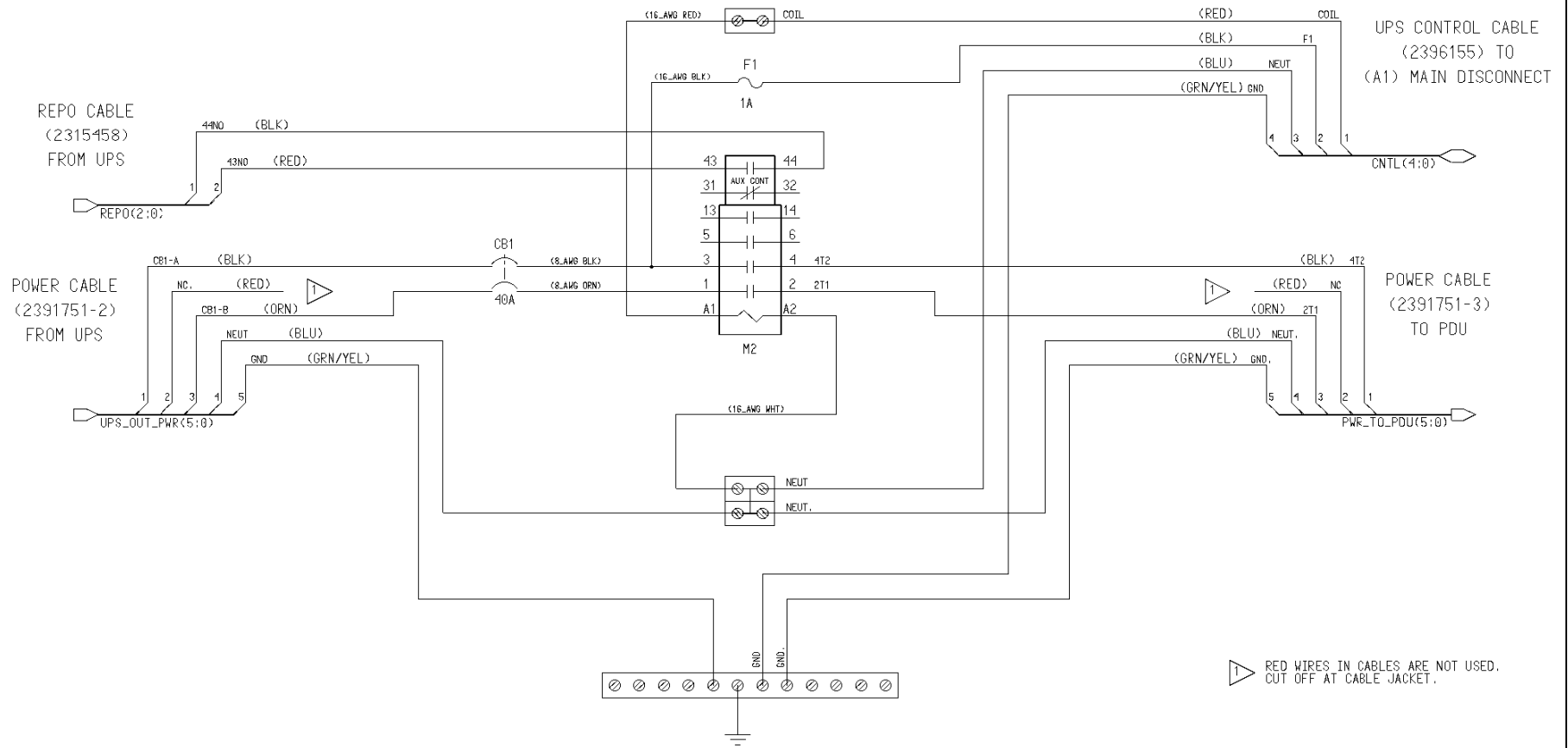


Figure 6-5 2391780 SCH, Sheet 5, Rev 0

Appendix A

Applicable GEMS Documents

The following documents provide reference information for the UPS Interconnect Kit.

Section 1.0

Reference Documents

- 2391780SCH UPS Interconnect Schematic
- 2326492SCH NGPDU Schematic
- 2391993PSP Purchase Specification for A1 Main Disconnect & UPS Control Panel
- 2388728PSP Purchase Specification for the UPS Disconnect Panel
- 2388733ADW Assembly Specification for the UPS Hardware Kit
- 46-015900 Global Documentation Standard
- 46-131001 Applied Practices (Assemblies)
- 46-170700 Applied Practices (Portions)

Section 2.0

Standards Documents

- 46-319010 Revision 0 Safety - Electrical power grounding 806-1
- 46-319011 Revision 0 Safety - Electrical leakage currents 806-2
- 46-319012 Revision 0 Safety Checklist for Design 806-3
- 46-319013 Revision 1 Operator Manual Safety/Regulatory Checklist 806-4
- 46-319015 Revision 0 Safety - Product Safety Labels and Instructions 806-6

Section 3.0

Procedure Documents

- POWERWARE 9150 User's Guide

Section 4.0

Government, Safety and Regulatory Documents

- UL 2601 General Requirements for Safety
- UL 94 Tests for Flammability of Plastic Materials
- CSA 22.2 Canadian Standards Association
- IEC 601-1 General Requirements for Safety
- EN 60601-1-2 IEC Normalized EMC specification
- IEC 801-2 Electro-Static Discharge
- IEC 801-3 Radiated Electro-Magnetic Fields Requirements
- IEC 801-4 Electrical Fast Transient / Burst Requirements
- IEC 801-5 Surge Immunity Requirements
- IEC 801-6 Immunity to Conducted Disturbances, Induced by RF fields above 9 kHz.
- IEC 1000-4-8 Magnetic Field

Section 5.0

Industry Standards

- NFPA-70 (1999) National Electrical Code (USA)

Appendix B

Field Replaceable Unit (FRU) Listing

The following listing identifies the Field Replaceable Units (FRUs) associated with this assembly. Vendor part numbers are included as examples only. For the complete list of approved vendors for each item, see the associated ESL on GEMS GPCP system.

GEMS PART #	NAME /DESCRIPTION	VENDOR	VENDOR P / N
2139965	UPS	POWERWARE	PF1221114UE0000
2388728	UPS Disconnect Panel	GE Supply	2388728
2388733	UPS Disconnect Panel	GE Supply	2388733

Table B-1 Field Replaceable Units

Appendix C

LED Panels

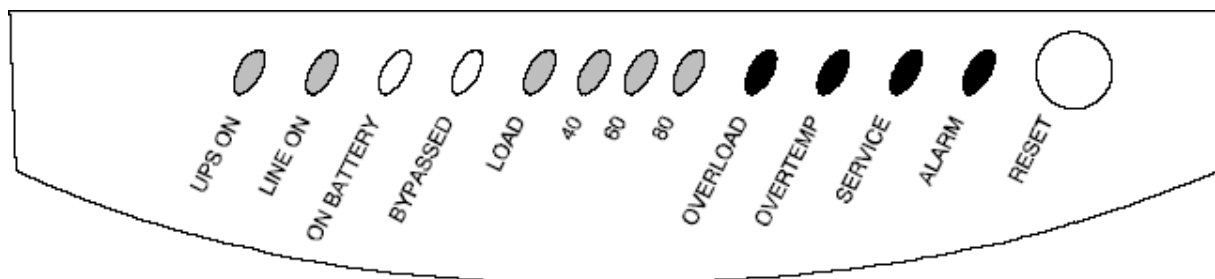


Figure 15. LED Panel

The following table shows the LED status and description.

LED	Status	Description
UPS ON	On	UPS is operating normally.
	Blinking	UPS is starting up or is shut down and waiting for power to return.
	Off	UPS is turned off and will not turn on automatically.
LINE ON	On	The utility voltage is acceptable for UPS and BYPASS operation.
	Off	The utility voltage is not acceptable for BYPASS operation. The UPS may still be operating on utility power.
ON BATTERY	On	UPS is in Battery Mode.
	Blinking	The battery voltage is low and has less than five minutes of backup time left before UPS shutdown.
BYPASSED	On	UPS is in Bypass Mode.
LOAD	On	Output is on.
	Off	No output.
40	On	> 40% load
60	On	> 60% load
80	On	> 80% load
OVERLOAD	On	Power requirements exceed the UPS capacity (greater than 100% of nominal).
	Blinking	Overload condition caused the UPS to shut down.

LED	Status	Description
OVERTEMP	On or Blinking	One of the main UPS components is too hot or the fan has failed.
SERVICE	Blinking	Battery failure, fan failure, or configuration error exists. Contact your service representative. See "Service and Support" on page 57.
ALARM	Blinking	There is a UPS alarm condition. See Chapter 6, "Troubleshooting" on page 55 for additional information.

Appendix D

Troubleshooting

The Powerware 9150 is designed for durable, automatic operation and also alerts you whenever potential operating problems may occur. Usually the alarms shown by the control panel do not mean that the output power is affected. Instead, they are preventive alarms intended to alert the user. Use the following troubleshooting chart to determine the UPS alarm condition.

Indicator or Alarm	Possible Cause	Action
ON BATTERY	LED is on. A utility failure has occurred and the UPS is in Battery Mode. The alarm sounds for five seconds when battery operation is initialized and then twice every minute.	None. The alarm resets when the condition becomes inactive.
	LED is blinking. The battery voltage is low and has less than five minutes of backup time left before UPS shutdown.	Prepare your equipment for shutdown. When utility is restored, the UPS restarts automatically, provides power to the load, and charges the battery.
BYPASSED 2 short beeps	The UPS is in Bypass Mode.	None. The alarm resets when the condition becomes inactive. Contact your service representative if the condition persists.
OVERLOAD	LED is on. Power requirements exceed the UPS capacity (greater than 100% of nominal).	Reduce the load. Your UPS continues to operate, but may switch to Bypass Mode if the load increases (112.5 - 150% of nominal for 30 seconds). The alarm resets when the condition becomes inactive.
	LED is blinking. Overload condition caused the UPS to shut down (greater than 150% of nominal for 0.3 seconds).	Reduce the load and restart the UPS. You may need to obtain a larger capacity UPS.
OVERTEMP On or Blinking	One of the main UPS components is too hot or the fan has failed.	Turn the Maintenance Bypass switch to the SERVICE position. Clear vents. Remove any heat sources. Allow the UPS to cool. If the alarm disappears, turn the Maintenance Bypass switch back to the UPS position. Contact your service representative if the condition persists.
	7 short and 3 long UPS beeps indicate the temperature inside the UPS is more than 50 degrees.	
	4 short and 4 long UPS beeps indicate a fan failure.	

Indicator or Alarm	Possible Cause	Action
ALARM On or Blinking	1 short and 4 long beeps indicate a neutral fault.	Input Neutral must be connected. Contact qualified service personnel (such as a licensed electrician).
	10 short and 5 long beeps indicate a neutral fault at startup.	
	7 short and 1 long beep indicates a neutral fault at shutdown.	
ALARM blinking 3 short beeps and 1 long beep.	REPO initiated.	None. The UPS remains off until the REPO switch is turned back on.
	The circuit connected to the REPO connector is open and preventing UPS operation.	Check that the REPO connector is in place and that the circuit connected to it forms a closed loop. Contact your service representative if the condition persists.
ALARM On or Blinking 5 short beeps and 4 long beeps.	Battery failure. Battery voltage dropped too fast and too low after charging.	Verify the battery circuit breaker(s) are turned on. If the condition persists, the batteries may need replacing. Contact your service representative.
ALARM On or Blinking 11 short beeps and 5 long beeps.	Input phase error.	Input phases must be rotated. Contact qualified service personnel (such as a licensed electrician).

Silencing the Alarm

Press the RESET pushbutton on the LED panel to silence and reset the alarm.



GE MEDICAL SYSTEMS

***3000 N, GRANDVIEW BLVD,
WAUKESHA, WI 53188, USA.***